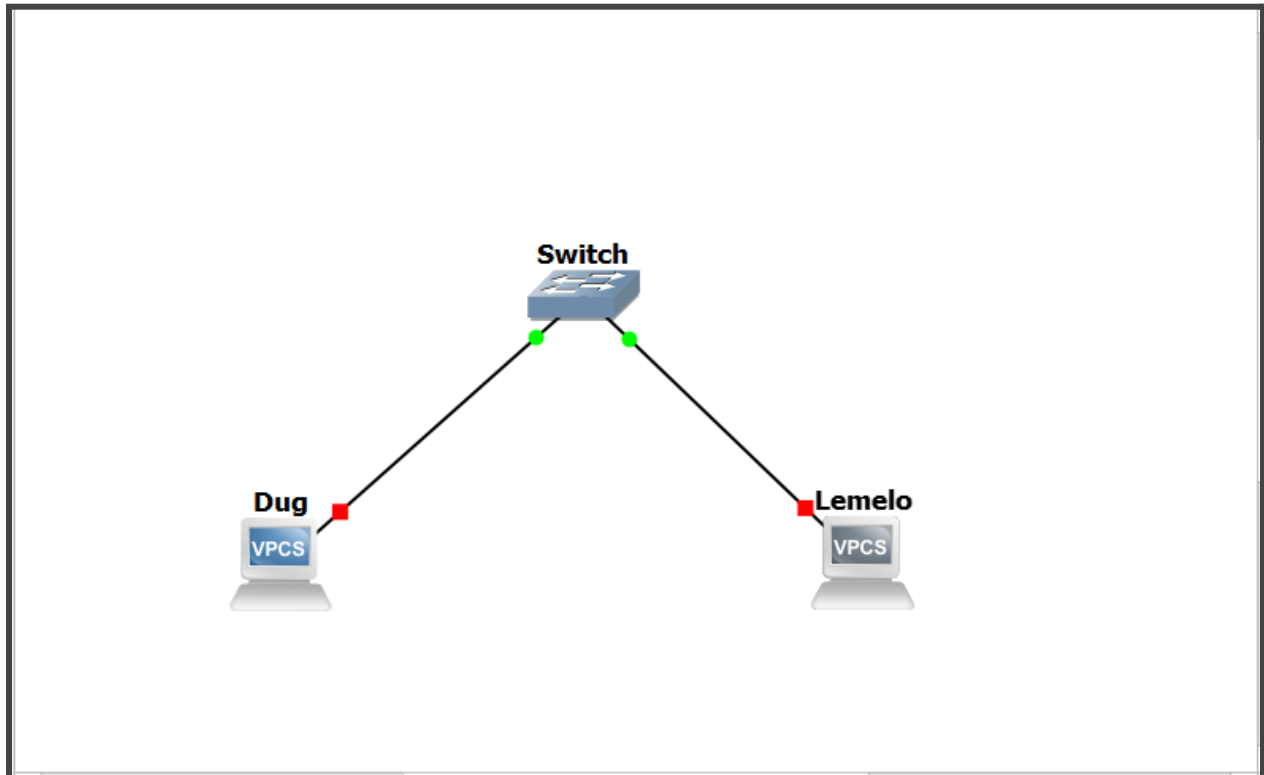




Part 1:-



Setting IP addresses to Dug (PC1) and Lemelo (PC2)

```
Dug> ip 192.168.1.1/24
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0
```

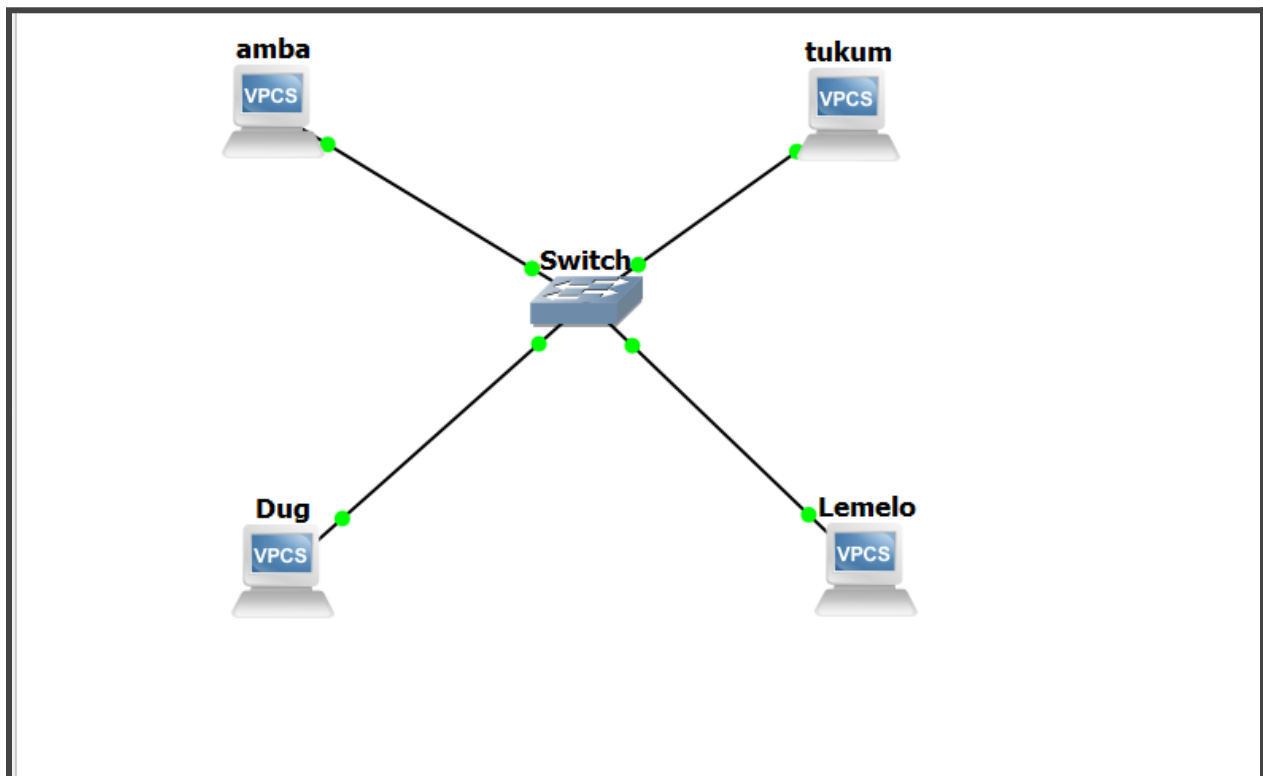
```
Lemelo> ip 192.168.1.2/24
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0
```

Pinging Dug from Lemelo and Lemelo from Dug

```
Dug> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.840 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.490 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.460 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=1.320 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.363 ms
```

```
Lemelo> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=1.112 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.070 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.373 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.407 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.085 ms
```

Part 2:-



Setting IP addresses to amba (PC3) and tukum (PC4)

```
amba> ip 192.168.1.129/25
Checking for duplicate address...
PC1 : 192.168.1.129 255.255.255.128
```

```
tukum> ip 192.168.1.130/25
Checking for duplicate address...
PC1 : 192.168.1.130 255.255.255.128
```

Pinging tukum from amba and amba from tukum

```
amba> ping 192.168.1.130
84 bytes from 192.168.1.130 icmp_seq=1 ttl=64 time=1.752 ms
84 bytes from 192.168.1.130 icmp_seq=2 ttl=64 time=1.270 ms
84 bytes from 192.168.1.130 icmp_seq=3 ttl=64 time=1.107 ms
84 bytes from 192.168.1.130 icmp_seq=4 ttl=64 time=1.331 ms
84 bytes from 192.168.1.130 icmp_seq=5 ttl=64 time=1.111 ms
```

```
tukum> ping 192.168.1.129
84 bytes from 192.168.1.129 icmp_seq=1 ttl=64 time=1.282 ms
84 bytes from 192.168.1.129 icmp_seq=2 ttl=64 time=1.857 ms
84 bytes from 192.168.1.129 icmp_seq=3 ttl=64 time=1.089 ms
84 bytes from 192.168.1.129 icmp_seq=4 ttl=64 time=1.030 ms
84 bytes from 192.168.1.129 icmp_seq=5 ttl=64 time=1.828 ms
```

Setting IP addresses and changing mask to Dug (PC1) and Lemelo (PC2)

```
Dug> ip 192.168.1.1/25
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.128
```

```
Lemelo> ip 192.168.1.2/25
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.128
```

Pinging Dug from Lemelo and Lemelo from Dug using new mask

```
Dug> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.616 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.734 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.321 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=1.756 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.110 ms
```

```
Lemelo> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=2.162 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=1.370 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=1.353 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=1.262 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=1.322 ms
```

Trying to ping tukum from Lemelo

```
Lemelo> ping 192.168.1.130
No gateway found
```

→ The reason why we get “No gateway found” is because Lemelo (PC2) and amba (PC3) are in different subnets. PC1 and PC2 have a valid host address of 192.168.1.1 - 126. What this means is that only IP addresses within this range can communicate with IP addresses within the range and nothing above or less.

While amba (PC3) and tukum (PC4) are in a valid host range of 192.168.1.129 - 254. And hence cannot ping PC1 or PC2.

This is all because we have a subnet mask of 255.255.255.128 /25 which creates two subnets.

Part3 :-

Configuring switch to split traffic into two different VLANS

Switch configuration

General

Name:

Switch

Console type:

none

Settings

Port:

8

VLAN:

1

Type:

access

QinQ EtherType:

0x8100

Add

Delete

Ports

Port	VLAN	Type	EtherType
0	1	access	
1	1	access	
2	1	access	
3	2	access	
4	2	access	
5	1	access	
6	1	access	
7	1	access	

Configuring VPCS to all be in the same subnet

```
Dug> ip 192.168.1.1 255.255.255.0
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0
```

```
Lemelo> ip 192.168.1.2/24
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0
```

```
amba> ip 192.168.1.3/24
Checking for duplicate address...
PC1 : 192.168.1.3 255.255.255.0
```

```
tukum> ip 192.168.1.4/24
Checking for duplicate address...
PC1 : 192.168.1.4 255.255.255.0
```

pinging PC2 from PC1 works but PC1 to PC3 doesn't

```
Dug> ping 192.168.1.2
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.153 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.282 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=1.173 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=1.399 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.526 ms

Dug> ping 192.168.1.3
host (192.168.1.3) not reachable
```

pinging PC4 from PC3 works but PC3 to PC2 doesn't

```
amba> ping 192.168.1.4
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=1.708 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=1.342 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=1.098 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=1.528 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=0.970 ms

amba> ping 192.168.2
No gateway found

amba> ping 192.168.1.2
host (192.168.1.2) not reachable
```

Task 4:-

Configuring VPCS to all be in the same subnet

```
Dug> ip 192.168.1.1/24
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0
```

```
Lemelo> ip 192.168.1.2/24
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0
```

```
amba> ip 192.168.1.3/24
Checking for duplicate address...
PC1 : 192.168.1.3 255.255.255.0
```

```
tukum> ip 192.168.1.4/24
Checking for duplicate address...
PC1 : 192.168.1.4 255.255.255.0
```

Ping from VPC1 to VPC2

```
Dug> ping 192.168.1.2 -t
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=1.159 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=1.140 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=2.106 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.889 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.023 ms
84 bytes from 192.168.1.2 icmp_seq=6 ttl=64 time=1.189 ms
84 bytes from 192.168.1.2 icmp_seq=7 ttl=64 time=2.103 ms
84 bytes from 192.168.1.2 icmp_seq=8 ttl=64 time=1.064 ms
84 bytes from 192.168.1.2 icmp_seq=9 ttl=64 time=1.915 ms
```

Wireshark capture of ICMP from both the VPCS

30 14.312749	192.168.1.2	192.168.1.1	ICMP	98 Echo (ping) reply	id=0x51d1, seq=30/7680, ttl=64 (request in 29)
31 15.330588	192.168.1.1	192.168.1.2	ICMP	98 Echo (ping) request	id=0x52d1, seq=31/7936, ttl=64 (reply in 32)
32 15.330588	192.168.1.2	192.168.1.1	ICMP	98 Echo (ping) reply	id=0x52d1, seq=31/7936, ttl=64 (request in 31)
33 16.350396	192.168.1.1	192.168.1.2	ICMP	98 Echo (ping) request	id=0x53d1, seq=32/8192, ttl=64 (reply in 34)
34 16.352396	192.168.1.2	192.168.1.1	ICMP	98 Echo (ping) reply	id=0x53d1, seq=32/8192, ttl=64 (request in 33)
35 17.367077	192.168.1.1	192.168.1.2	ICMP	98 Echo (ping) request	id=0x54d1, seq=33/8448, ttl=64 (reply in 36)
36 17.369602	192.168.1.2	192.168.1.1	ICMP	98 Echo (ping) reply	id=0x54d1, seq=33/8448, ttl=64 (request in 35)
37 18.399572	192.168.1.1	192.168.1.2	ICMP	98 Echo (ping) request	id=0x55d1, seq=34/8704, ttl=64 (reply in 38)
38 18.400572	192.168.1.2	192.168.1.1	ICMP	98 Echo (ping) reply	id=0x55d1, seq=34/8704, ttl=64 (request in 37)
39 19.410618	192.168.1.1	192.168.1.2	ICMP	98 Echo (ping) request	id=0x56d1, seq=35/8960, ttl=64 (reply in 40)

→ The reason why we cant see any packets captured on PC3 is because we are only sending a ping request from PC1 to PC2 and a switch is being used to send the request and not a hub.

→ If it was the other way around and used a hub then all the PCs would get the request and we would see all the PCs getting ICMP request.